

NATHANIEL A. RODGERS

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EDUCATION

Columbia University
Bachelor of Arts, Astrophysics
GPA: 3.91 | Dean's List

New York, NY
Anticipated May 2027

- **Relevant Coursework:** Applied Quantum Mechanics; Quantum Mechanics; Advanced Mechanics; Advanced E&M; Astrophysical Magnetism; Quantum Computing & Simulation

PROFESSIONAL EXPERIENCE

Columbia University, Undergraduate Research Fellow | Columbia University May – September 2026

- Applied hierarchical Bayesian inference via NumPyro, JAX, and TensorFlow to execute pulsar parameter estimation, constraining neutron star ellipticities to optimize LIGO gravitational wave search mechanisms.
- Developed Python workflows to process tabular CSV/Excel data, independently implementing model constraints derived from academic literature reviews.
- Secured and managed summer funding by authoring technical grant requests while collaborating with a Columbia University mentor to drive project milestones.

Math, Physics, & Computer Science Tutor | New York, NY Sept 2021 – Present

- Tutors Columbia University students in Calculus I-IV, Physics I & II, and Data Structures, emphasizing problem solving

Columbia University, Research Assistant | Columbia University February – September 2025

- Automated processing of datasets exceeding 1 million entries using Python and shell scripts for stellar evolution and merger research, reducing analysis turnaround time by hundreds of hours
- Adapted and executed stellar evolution code (MESA) to simulate single, binary, and merging stars across a multi-dimensional parameter space, testing tens of variables along five axes
- Compiled technical presentations translating advanced stellar astrophysics for the research team, guiding project decisions

Write on Sports, Intern | South Orange, NJ July 2024

- Taught journalism techniques to a class of 30 early high school aged students
- Motivated students to ideate, ask meaningful questions, and engage in substantial conversations around topics of interest
- Provided lessons on conducting interviews, effective note-taking, and writing newspaper-style feature and spot stories

ACTIVITIES

Columbia Space Initiative, Mission Lead | Columbia University Sept 2023 – Present

- Leads development of high-altitude balloon systems reaching 120,000+ ft in continuous flight conditions
- Scaled team from 2 to 6+ engineers, overseeing project management, task delegation, and technical mentorship
- Directs end-to-end design, fabrication, and testing of scientific payload systems for near-space environments
- Spearheading active development of a 72+ hour autonomous ballast system capable of operating at -60°C , enabling long-duration flight through extreme atmospheric conditions
- Designs and integrated embedded control systems incorporating multi-sensor feedback (4+ sensors), actuators, GPS, and satellite communications
- Develops custom PCBs and electronics using KiCad; programming embedded systems in C and building supporting tools in Python
- Utilizes CAD, machining, and laser cutting to manufacture flight-ready hardware from scratch

SELECTED PROJECTS

Experimentally Testing Bell's Inequality and Single Photon Tomography April – May 2026

- Confirmed the quantum mechanical violation of the Clauser-Horne-Shimony-Holt inequality using an optical table and polarization entangled photon pairs created via spontaneous parametric down-conversion
- Realized measurement correlation parameter $S = 2.297$, greater than the classical limit of $S=2$, proving quantum mechanics
- Probed the system's coherence by performing quantum-state tomography on the constituent photons, recovering full density matrices of each individual photon

Designing and Installing a Fiber-fed Spectrograph for Multi-Object Spectroscopy March – May 2026

- Designed, fabricated, and installed a plug plate, optical fibers, and spectrograph on the Pupin Rutherford Dome telescope
- Mapped celestial field onto plug plate through a gnomonic projection, cut, measured, and polished 100 micron fibers with ferrule attachments, and transmitted to an optical table spectrograph consisting of collimator, grating, and focusing lenses
- Calibrated multi-fiber spectrograph by writing a Python data reduction pipeline to isolate staggered fiber tracks, map pixel coordinates to precise wavelengths using neon emission lines, and model a uniform $\sim 8.5\text{ nm/mm}$ linear dispersion relation

COMPUTER SKILLS

Python; Java; C++; MATLAB; CAD; KiCad; Excel (X-Lookup, Pivot Tables); Word; PowerPoint; Google Workspace; Outlook; Slack